V-Trace: Demo & Assessor Guide

How to run, sign in and try V-Trace

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This guide has three parts: how to start V-Trace, a scripted demo for the presentation (with the results you should see), and things an assessor can try for themselves.

1. Start the app

Option A – hosted (no install). Open the Streamlit Community Cloud link shared with the submission and sign in with the assessor account provided with it. The four demo cameras load automatically.

Option B – local (Windows, Python 3.11).

1. `python -m venv .venv` and `.venv\Scripts\pip install -r requirements.txt`
2. Copy `.streamlit/secrets.toml.example` to `.streamlit/secrets.toml` and set a username and password.
3. `.venv\Scripts\python -m streamlit run app.py`, open http://localhost:8501 and sign in.

The app has three pages (left sidebar): **Search**, **Cameras** and **How it works**. It opens in dark mode; switch with ⋮ (top right) → Settings → Theme.

2. Presentation demo (about 4 minutes)

The videos (sidebar → *Demo videos*, already processed). `camA_west`, `camB_centre` and `camC_east` are three cameras along one street, so the same cars pass A → B → C (near lane) or C → B → A (far lane). `camD_other_road` is a different road whose vehicles never appear in A–C.

Real-world time and location. Every camera has a location and a recording start (**Cameras** page); a sighting's real-world time = recording start + its position in the video. The demo videos carry no such metadata, so their details are curated and placed in Singapore: camA–camC on Orchard Road around the Scotts Road junction, all starting **Wed 28 Oct 2026, 08:15:00** (same clock), and camD on Bukit Timah Road near Newton Circus, starting **08:17:30** (2.5 min later).

The Search page has three steps: **1 Describe the vehicle**, **2 Best matches**, **3 Where this vehicle was seen**.

Scenario A – find a red car and trace it through three cameras (main demo)

1. **1 Describe the vehicle**: Colour **red**, Type **suv**, click **Search**.
2. **2 Best matches**: #3 is a red Citroën in **camB_centre, 08:15:42.0 – 08:15:43.5**. Click **Select**.
3. *Expect:* **3 Where this vehicle was seen** shows **Route: camA_west 08:15:40.8 → (+1.2 s) camB_centre 08:15:42.0 → (+1.6 s) camC_east 08:15:43.5**, with one row per camera: the location, the real-world date and time span, the time in the video, a crop and the full frame with the car boxed, plus a map of the camera locations. camA scores 0.67 and camC 0.51. **camD_other_road** is listed under *Not seen in* ("no vehicle within 10.0 s"): it was recorded minutes later.

Scenario B – the same search, an ambiguous result

- In the same results, select #2 (the same red Citroën in **camA_west, 08:15:40.8**).
- *Expect:* camB_centre and camC_east are listed under *Not seen in*, each with its three closest vehicles. In camB the true car (08:15:42.0) is first with 0.67, but another red car (08:15:39.9) scores 0.62, too close to call, so the ratio test leaves the choice to the user. Good talking point: the system shows evidence, it does not claim an identity. (Searching from the camB sighting, as in Scenario A, gives a clear answer because camB's view of the car is sharper.)

Scenario C – real-world time decides (cameras a minute apart)

- **Cameras** page: change camC_east's recording start to **08:14:00** (1 min earlier) and click **Save camera details**.

- Repeat Scenario A: camC's sighting is now at 08:14:43.5, about a minute before the others, so with the 10 s limit it is *Not seen*. **Cameras** page → *Matching settings* → *Max time between cameras* **1 min 00 s**: it is matched again and the route is re-ordered in real-world time (camC → camA → camB). camD (recorded from 08:17:30) stays out. Set camC back to 08:15:00 afterwards.

Scenario D – upload videos (test pack)

- Sidebar → **Upload new videos**, add the three `set1_orchard_*.mp4` files from `test_videos/` (optionally also `set3_bukit_timah.mp4`) and click **Process videos** (about 20–25 s per clip on a laptop). The location and start time fill in automatically for these files, and the real-world time limit is switched on.
- Describe Colour **gray**, Type **suv**, select **#1** (west camera, 17:45:04.0): the route shows the Scotts Road junction 3.2 s later. More scenarios with expected results are in `test_videos/README.md`.
- For your own clips, open each file's *location & start time* box and enter them before processing; for uploads with unknown clocks the real-world time limit stays off (**Cameras** page → *Matching settings*). Details can be edited later on the **Cameras** page.

Talking points while it runs

- Every model is pretrained and used as published: YOLO11 + ByteTrack (sightings), CLIP zero-shot (description), EasyOCR (plates), and the FastReID VeRi-776 Re-ID model from the VeRi authors' team (same vehicle across cameras).
- A sighting = all frames in which one vehicle is seen continuously in one camera; its real-world time = camera recording start + its position in the video, so cameras can be seconds, minutes or hours apart.
- Step 1 uses only the fields you fill in; a missing field is ignored rather than counted as a mismatch.
- A camera counts as "seen" only when its best sighting is similar enough (≥ 0.40), clearly better than the next one and, with the time limit on, close enough in real-world time (default 10 s).
- **How it works** page: the models and the VeRi-776 benchmark numbers.

3. Search combinations that work

Every combination below was checked against the reference identities (the full-frame tracking used in `scripts/demo_eval.py`). Leave all other fields empty, click **Search**, then **Select** the listed result. Settings at their defaults (threshold 0.40, real-world time limit on, 10 s).

Built-in demo cameras (sidebar → *Demo videos*)

Describe	Select	Where this vehicle was seen
Colour **red**, Type **suv**	#3 · camB_centre · 08:15:42.0	camA 08:15:40.8 → camB 08:15:42.0 → camC 08:15:43.5 (all three cameras)
Colour **green** (Type any)	#2 · camA_west · 08:15:00.9	camA 08:15:00.9 → camB 08:15:06.6 → camC 08:15:07.9 (all three cameras)
Colour **black**, Type **suv**	#1 · camA_west · 08:15:02.8	camA 08:15:02.8 → camB 08:15:05.7
Colour **black**, Type **suv**	#3 · camA_west · 08:15:20.0	camB 08:15:18.2 → camA 08:15:20.0 (far lane, B before A)
Colour **white**, Type **sedan**	#3 · camA_west · 08:15:22.4	camB 08:15:20.5 → camA 08:15:22.4 (far lane)

The green car's camA track briefly switches to another car (one crop shows it): a good talking point for why the crops are shown.

Test pack, set 1 (upload the three `set1_orchard_*.mp4`, optionally with `set3_bukit_timah.mp4`)

Describe	Select	Where this vehicle was seen
Colour **gray**, Type **suv**	#1 · west · 17:45:04.0	west 17:45:04.0 → Scotts junction 17:45:07.2
Colour **green**, Type **sedan**	#1 · west · 17:45:02.2	west 17:45:02.2 → Scotts junction 17:45:08.2
Colour **white**, Type **suv**	#4 (#7 with set 3) · Scotts junction · 17:45:17.8	west 17:45:17.0 → Scotts junction 17:45:17.8
Colour **white**, Type **suv**	#6 (#9 with set 3) · west · 17:45:01.0	west 17:45:01.0 → Scotts junction 17:45:09.2

Test pack, set 2 (upload the three `set2_orchard_*.mp4`, optionally with `set3_bukit_timah.mp4`)

Describe	Select	Where this vehicle was seen
Colour **red**, Type **suv**	#3 · Scotts junction · 17:45:43.2	west 17:45:37.6 → Scotts junction 17:45:43.2
Colour **gray**, Type **suv**	#2 · west · 17:45:25.4	Scotts junction 17:45:23.8 → west 17:45:25.4
Colour **gray**, Type **bus**	#2 · west · 17:45:36.2	west 17:45:36.2 → Scotts junction 17:45:37.4
Type **bus** (Colour any)	#4 · west · 17:45:36.2	same bus as above

In the test sets the east camera is listed under *Not seen in* with its closest vehicles: with *Fast processing* the car is seen there too briefly to keep, or no candidate is a confident match.

Test pack, set 3 (Bukit Timah Road, uploaded together with set 1 or 2)

Describe	Top Bukit Timah result	Where this vehicle was seen
Colour **black**, Type **sedan**	Bukit Timah · 17:48:02.0	only Bukit Timah Road; Orchard Road cameras *Not seen*
Colour **white**, Type **van**	Bukit Timah · 17:48:01.2	only Bukit Timah Road
Type **bus**	Bukit Timah · 17:48:11.6	only Bukit Timah Road. Turn off *Limit by real-world time* (**Cameras** page) and it is wrongly matched to two Orchard Road cameras: this shows why the time limit matters

Combinations to avoid in a live demo (checked: the suggested match is a different vehicle): demo Colour gray + Type suv #4, Colour white + Type van, Colour green #4.

4. Things to try (assessors)

Try this	Where	What it shows
Search with only a colour, then add the type	Search 1	each extra field narrows the list
Type a partial plate with wildcards, e.g. `SGX1?3?`	Search 1	wildcard plate matching (plates are rarely readable in the demo footage, so unread plates get a neutral score)
Pick a distinctive feature or type free text, e.g. "bus"	Search 1	CLIP text–image scoring
Select different vehicles from the results	Search 3	per-camera sightings, or the closest candidates when there is no confident match
Change the threshold, the *Max time between cameras* or turn off *Limit by real-world time*	Cameras page → Matching settings	stricter / looser matching; the real-world time window
Edit a camera's location or recording start	Cameras page	real-world times and the route update
Upload new videos with a set from `test_videos/` (see its README)	sidebar	the full pipeline on new footage, with expected results

Known limitations to keep in mind: plates are not readable at this camera distance; colour labels are unreliable for silver/grey cars; a car seen very small or from a very different angle may not be matched; results are ranked candidates for human review, never identity claims.